Tran Dang Trung Duc

Open to Senior AI Engineer / AI Researcher

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Experience

•VNGGames – AI Engineer *05/2025 - present, Ho Chi Minh City*

- **AI-powered Art Production Pipeline (for Game Art Team):** Designed and developed a standardized **AI-driven 2D & 3D asset production pipeline** for game development, integrating **3D generation, texture synthesis, retopology**, and automated **2D workflows** (image upscaling, generation, editing) with prompt templates and visualization tools. Built a prototype system for **AI-powered in-game image generation with content moderation**, enabling **personalized player experiences** through **high-speed generative models**. Fine-tuned and integrated multiple **open-source diffusion and generative models** to support advanced art workflows such as **live painting, inpainting, image enhancement, style reference, and ControlNet-based control**. Developed an internal **AI Portal** supporting **4,000+ prompt templates** and **custom automated workflows** for scalable game asset generation. Implemented **MCP servers** to connect **Claude Desktop/Cursor with Blender, Maya, and the AI Portal**, enabling **prompt-driven automation of 3D asset pipelines**. *Team size:* 2 *(Role: Team Leader).*
  Tech Stacks: Python, Docker, Diffusion Models (**Hunyuan3D, Era3D, Sana, SDXL, Flux, Qwen Image**), VAR Models (**Infinity**), AI Guardrails, ComfyUI, Automatic1111, Claude, Gemini, MCP, TypeScript, React, Vercel, Blender, Maya, Krita. Demo: *Link*.
  Achievement: Achieved a **40% reduction in game asset production time** and **validated the feasibility of integrating generative AI directly into large-scale game production workflows**.
- **AI Companion Character (for PM & Department Head)**
  Description: Developed an interactive **AI Companion character** integrating **LLMs**, **Text-to-Speech (TTS)**, **Automatic Speech Recognition (ASR)**, and **Live2D** technologies, enabling **real-time multimodal interaction** including dialogue, lipsync, facial expressions, and motion control. *Team size:* 1
  Tech Stacks: Python, TypeScript, **LLMs (Qwen, Gemini)**, **ASR (Sherpa-onnx)**, **TTS (IndexTTS2)**, Live2D. Demo: *Link*
  Achievement: Built **two functional PoC systems** demonstrating **real-time multimodal AI interaction**, validating the concepts of **AI VTuber** and **AI Monitoring systems**.
- **Building a RAG System (for Game Design & QA/QC):** Led the design and development of a **NotebookLM-like Multimodal RAG system** over **Game Design Documents (GDDs) and Game Source Code** to support **QA/QC automation, project validation, and design–implementation consistency analysis**. The system enables intelligent retrieval and contextual reasoning across complex technical documents used in game production. *Team size:* 3 *(Role: Team Leader).*
  Tech Stacks: Python, JavaScript, **LLMs (Qwen Turbo, Gemini, OpenAI-compatible)**, LanceDB, PostgreSQL, **Vector Databases (Supabase)**, Document Analysis & OCR (**marker-pdf**), Flask. Demo: *Link*.
  Achievement: Delivered a **NotebookLM-style multimodal RAG system** with **sub-3s retrieval latency**, capable of processing **complex document layouts** including **figures, flowcharts, and tables** for large-scale game development documentation.
- **AI Tank Bot using Reinforcement Learning (for Game Prototype):** Designed and developed an **AI-controlled tank agent** for a **Brawl Stars–style multiplayer tank shooter**, capable of **autonomous navigation, enemy detection, and combat decision-making**. The agent learns optimal behaviors through **Reinforcement Learning** within the Unity simulation environment. *Team size:* 1.
  Tech Stacks: Python, C#, Unity, **Unity ML-Agents**, Reinforcement Learning. Demo: *Link*.
  Achievement: Delivered a functional **Reinforcement Learning game agent** capable of **real-time navigation and combat strategy learning**, demonstrating the feasibility of applying **RL-based AI agents in action game environments**.
- **AI-integrated Game Prototypes with Industry Partners:** Collaborated directly with **Tencent and Alibaba** to explore and implement **AI-powered game development workflows**. Designed and developed two **AI-integrated game prototypes**: a **2D pixel-art game** built on **Alibaba Cloud's Qwen multimodal architecture**, and a **3D web-based interactive experience** leveraging **Tencent's Hunyuan3D generative technology**. These prototypes demonstrate the integration of **generative AI, voice synthesis, and real-time 3D generation** within playable game environments. *Team size:* 1.
  Tech Stacks: **Vision Models (Qwen Image, Wan 2.5/2.6)**, Voice (**Qwen TTS**), 3D Generation (**Hunyuan3D – image & animation**), Three.js, Pygame. Demo: *Link*.
  Achievement: Successfully delivered **two AI-driven game prototypes** demonstrating the feasibility of **integrating generative AI technologies into interactive game experiences**.

•MCI Academy – AI Instructor *05/2025 - 09/2025, Ho Chi Minh City*

- **First and core AI Instructor** at MCI Academy in Ho Chi Minh city. Designed and delivered corporate training courses on **AI Application**, **Computer Vision**, and **n8n**, including both content development and instruction.
- Achievement: **delivered** the **largest number of AI training programs** within the organization (**7 programs, 114 trainees, 45 sessions**), achieving an average **feedback rating of 4.70/5.0** for overall in **5 months**.

•Freelance & Contract AI Projects – AI Engineer *03/2025 - 05/2025, Ho Chi Minh City*

- **Contract AI Engineer** at Alchemy Technologies LLC, **developed a production-ready AI Agent** for **Intelligent Customer Service**, integrating **RAG (Vertex AI)**, Google Search, **dynamic Few-shot prompting**, and external tools (**web-browsing agent**). Deployed and monitored the system on **Google Cloud**.
  Tech Stack: Vertex AI, Google Agent Development Kit, RAG Engine, Docker, Gemini, ElevenLabs (TTS, STT, Voice Cloning).

- **Contract AI Engineer** at *Blue Dragon Communications LLC*, analyzed requirements from MakinaRocks and **designed an AI solution proposal** to **automate Construction Drawing Review**.
  Tech Stack: AWS, Docker, Document Layout Analysis (DocLayout-YOLO), OCR (marker-pdf, chandra-ocr, markitdown), Table Structure Recognition (TART, TC-OCR), Segmentation.
- **Freelance AI Trainer** at *Yen Lam Company LLC* and *Hoang Bao An Trading Production LLC*, **delivered corporate training** on **ChatGPT & Gemini applications** for enterprise staff.
- **Achievement: Google ADK Contributor** (since April 19). Received **2 Full-time AI Engineer Offers** from Alchemy Technologies LLC (**successfully piloted an AI Agent on Telegram & Zalo for 1 month**) and *Blue Dragon Communications LLC* (**project proposal approved and secured**).

•Seoul National University of Science and Technology – AI Researcher *09/2022-08/2025, Seoul*

- **Core member** working on a project of 3D Digital Media Streaming Service Technology and 4D Lidar Panoptic Segmentation in part by the National Research Foundation of Korea grant **funded by the South Korean government**.
- **Analyzing project** requirements (3D instance segmentation / 4D panoptic segmentation), **research existing AI methods** to find strengths and weaknesses, propose and **develop a novel solution** that ensures requirements with higher performance.
  Tech Stacks: Python, C++, Docker, Pytorch, Pytorch3D, Open3D, OpenCV, CUDA, KT Cloud.
- Achievement: **Top 1 LSTQ highest score** on the 4D Panoptic Segmentation Benchmark and **Top 2 mAP$_{25}$ highest score** on the 3D Semantic Instance Benchmark (**Conference rank A***) at submission time. **A patent** of application in South Korea. Winner of **Excellent Thesis Award** for Master Student.

•CBD Robotics Limited Liability Company – AI Internship *04/2021 - 10/2021, Ho Chi Minh City*

- Learned and practiced the **fundamentals of Machine Learning** (EDA, Feature Engineering, Hypothesis, Bagging, SVM, Clustering,...) and Deep Learning (Image classification, Topic modeling,...).
- Final Project: a rule-based combined with model-based (ParlAI) Chatbot for a laptop business, deployed on GCP.
- Achievement: **Top 1 highest project score** of the bootcamp.

•Bosch Global Software Technologies Company – Software Engineer *05/2021 - 08/2022, Ho Chi Minh City*

- Working on **automotive steering developer team** for the Chinese OEM (Geely, Cherry,...) and Japanese OEM (Toyota, Honda, Subaru,...)
- **Developing components and functions on demand**, first person to approach Jenkins (write code for integration test and retrain the team), Lab-testing, unit testing, requirement checking.
- *Tech Stacks:* C, Python, IBM ClearQuest, IBM ClearQuest, IBM ClearCase, Static Code Analyzer, Jenkins, and internal software.
- Achievement: **25%** increase in **work efficiency** after first 3 months of official employment, develop internal python-based tools to **reduce task** time by **2 times.**

Research Publication

[1] **Tran DTD**, Lee Y. Transformer-supported Tackling 3D Point Cloud Instance Segmentation. In **Proceedings of the 10th Annual Conference of Vietnamese Young Scientist** (pp. 40).
[2] **Tran DTD**, Kang B, Lee Y. MSTA3D: Multi-scale Twin-attention for 3D Instance Segmentation. In **Proceedings of the ACM International Conference on Multimedia 2024** (pp. 1467–1475).
[3] Park Y, **Tran DTD**, Kim M, Kim H, Lee Y. SP2Mask4D: Efficient 4D Panoptic Segmentation Using Superpoint Transformers. In **IEEE International Conference on Electronics, Information, and Communication (ICEIC) 2025** (pp. 1–4).
[4] Park Y, **Tran DTD**, Lee Y. SAFeR4D: Scene-Aware Feature Representation for Efficient 4D Panoptic Segmentation. **Under Review at IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)**. Aug 2025.

Education

•Master Program - Artificial Intelligence *09/2022-02/2025*
Department of Electrical and Information Engineering, SNUT, South Korea GPA: **4.45/4.5** with 33 credits

•Master Program - Data Science *11/2021-03/2022*
Department of Computer Science and Technology, HCMUT - VNUHCM, Vietnam GPA: **8.07/10** with 18 credits

•High-Quality Engineer Program (Equivalent to Master) - Mechatronics *09/2015-03/2016, 04/2017-11/2021*
Department of Mechanical Engineering, HCMUT - VNUHCM, Vietnam GPA: 7.87/10 with **268 credits**

•Japanese Preparatory Class of Government Scholarship - Japanese Language *04/2016 - 10/2016*
Department of Japanese, Hanoi Univeristy, Vietnam **N3 (Top 1/10)** and **N2 in 10 months**

Achievements & Additional Information

Scholarships: Awarded multiple academic scholarships including **75% tuition scholarship for all semesters** at **SNUT**, **Korean Professor Scholarship**, **Vietnamese Government Scholarship (Top 10 students)** for nuclear energy, **Vietnam–Japan Human Resource Training Scholarship** at **HCMUT**, **100% tuition scholarship (Top 2 students)** at **IPU**, and a **Full PhD Scholarship Offer** at **SNUT**.
Languages: Vietnamese (**Native**), English (**Professional**), Japanese (**JLPT N2**), French (**B1**).
Soft Skills: Problem-solving, Independent work, Team collaboration, Presentation skills, Strong work ethic.
Portfolio: Website | LinkedIn | GitHub | Scholar | HuggingFace | StackOverflow | Kaggle (3 Experts) | BrainHub AI Tools